

Creating an OData Service in SAP

1. Create a Data Dictionary (DDIC) Structure (You can Skip this step in case your fields from one table)
This structure will define the data model for your OData service.

Go to transaction SE11.

Select Data Type and click Create and choose structure.

The screenshot shows the 'ABAP Dictionary: Initial Screen' in SAP. The 'Data type' radio button is selected, and the text 'ZODATA_STR_68' is entered in the adjacent field. Other options like 'Database table', 'View', 'Type Group', 'Domain', 'Search help', and 'Lock object' are unselected. At the bottom, there are three buttons: 'Display', 'Change', and 'Create'.

Define the structure name and fields that your OData service will expose.

The screenshot shows the 'Dictionary: Change Structure' screen. The structure name 'ZODATA_STR_68' is entered, and the short description is 'Structure for Odata Service'. The 'Components' tab is active, showing a table of components. The table has columns for Component, Typing Method, Component Type, Data Type, Length, Decim..., Coordinate, Short Description, and Group. Three components are listed: MATNR (Material Number), MATKL (Material Group), and MATX (Material Description).

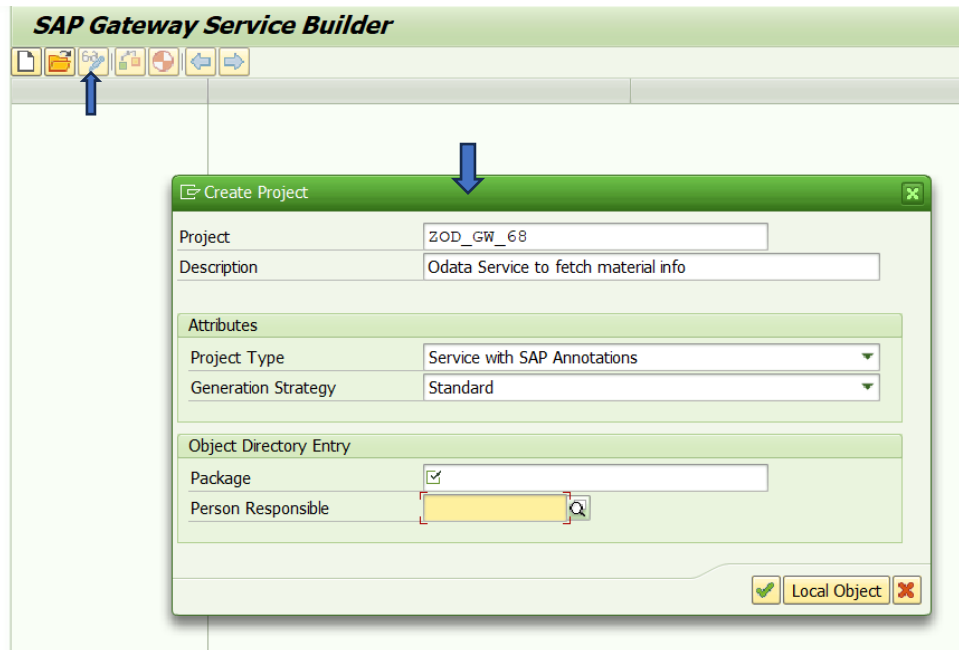
Component	Typing Method	Component Type	Data Type	Length	Decim...	Coordinate	Short Description	Group
MATNR	Types	MATNR	CHAR	40	0		0 Material Number	
MATKL	Types	MATKL	CHAR	9	0		0 Material Group	
MATX	Types	MATX	CHAR	40	0		0 Material Description	

2. Create a Gateway Project

Go to transaction “SEGW”

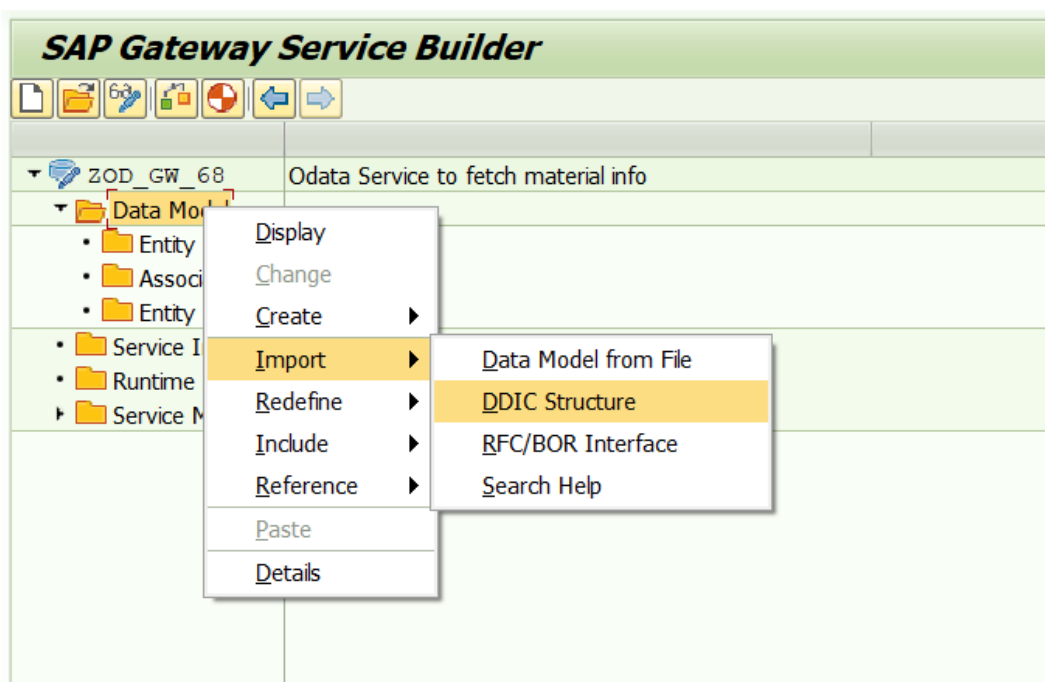
Click Create Project.

Give the project name (e.g., ZOD_GW_68) and enter the required details.



2.1 Define Data Model

Right Click on Data Model → Import → DDIC Structure



2.2 Give it any name you want

In ABAP structure : write the structure name we made in Step 1 (or table name as structure).Then Next

Wizard Step 1 of 3: Import from DDIC Structure

Create an Entity Type or Complex Type

Name: ODHEADER

☒ Entity Type ☐ Complex Type

Import from ABAP Structure

ABAP Structure: ZODATA_STR_68

☒ Create Default Entity Set

Next Cancel

Choose the field that you want to display in your Odata then Next

Wizard Step 2 of 3: Import from DDIC Structure

Select Parameter(s)

Data Source Parameter	Assign Structure	Description	Type	Length	Decimals	Import Sear...
☒ ZODATA_STR_68	☐		ZODATA_STR_68			☐
☒ MATNR	☐	Material	CHAR	40		☐
☒ MATKL	☐	Material Group	CHAR	9		☐
☒ MAKTX	☐	Description	CHAR	40		☐

Back Next Cancel

Choose a field key which is unique and click Finish.

Wizard Step 3 of 3: Import from DDIC Structure

Modify Entity Type

IsEnti...	Complex/Entity Type Name	ABAP Name	Is Key	Type	Name	Label
☒	ODHEADER	MATNR	☒	CHAR	Matnr	Material
☒	ODHEADER	MATKL	☐	CHAR	Matkl	Material Group
☒	ODHEADER	MAKTX	☐	CHAR	Maktx	Description

Back Finish Cancel

Now click Save, then click on “Generate Runtime Objects”.

SAP Gateway Service Builder

Messages

- Project 'ZOD_GW_68' has been checked; no errors were found

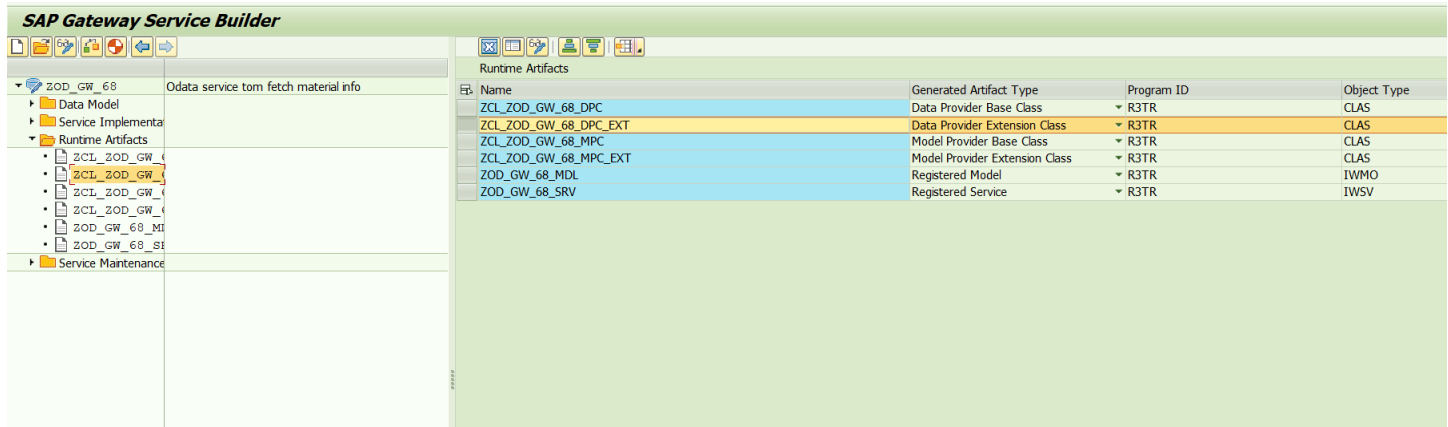
Nu... 1

Project 'ZOD_GW_68' has been checked; no errors were found

Continue to generate the required runtime classes and service objects.

Open ‘Runtime Artifacts’

- 1- Click on Class ‘ZCL_ZOD_GW_DPC_EXT’
- 2- Click on the same as shown in picture.

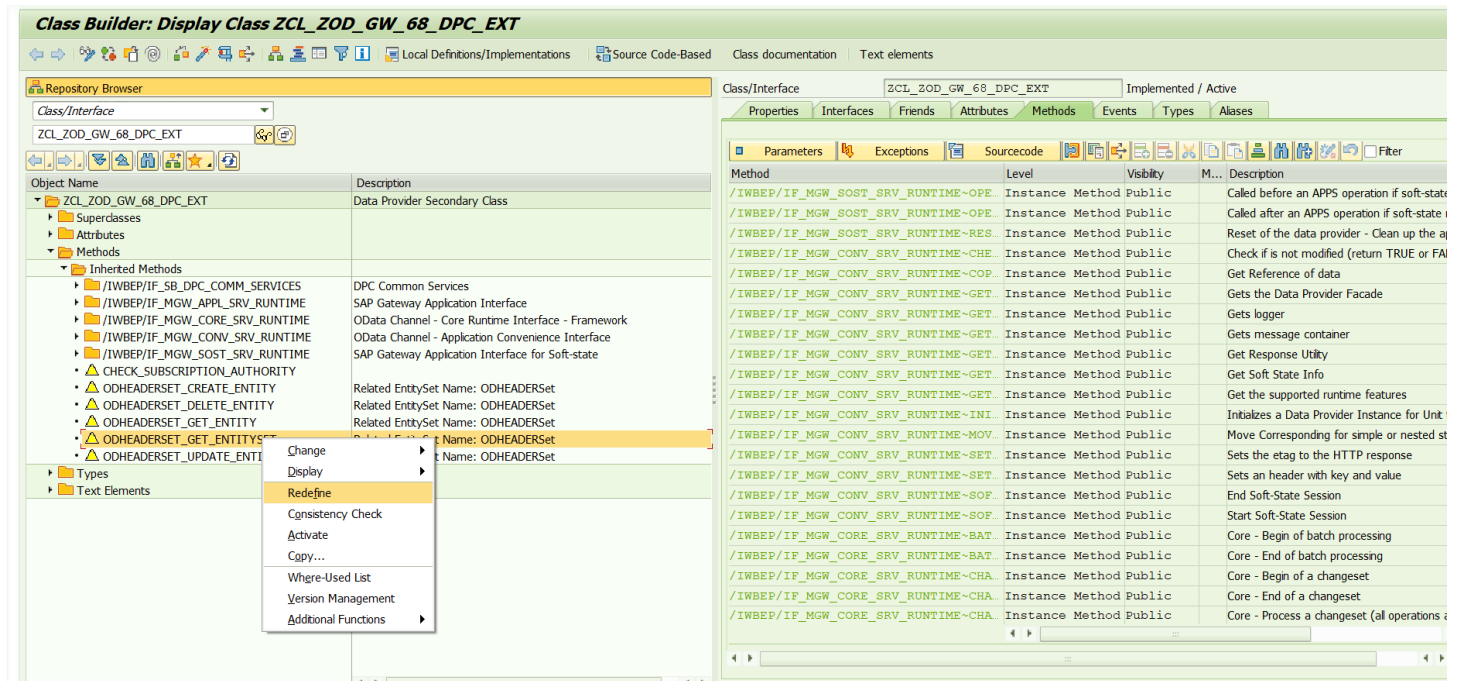


Methods → Inherited Method

It will show the CRUD operations: Create, Read, Update, and Delete.

We'll use GET this time

Right Click GET_ENTITYSET → Redefine.



After Redefine the selected operation, Here we write the code to fetch the data

Darshan Aher (SAP ABAP Developer)

Note: ET_ENTITYSET is the internal table which we store the data in
(it take same structure we use in step 1).

Activate the method as you would activate any ABAP program.

```
Method | ODHEADERSET_GET_ENTITYSET | active
1 | METHOD odheaderset_get_entityset.
2 |
3 |     SELECT mara~matnr, matkl, maktx
4 |     FROM mara
5 |     INNER JOIN makl ON makl~matnr = mara~matnr
6 |     INTO CORRESPONDING FIELDS OF TABLE @et_entityset
7 |     UP TO 10 ROWS.
8 |
9 | **TRY.
10 | *CALL METHOD SUPER->ODHEADERSET_GET_ENTITYSET
11 | * EXPORTING
12 | *     IV_ENTITY_NAME =
13 | *     IV_ENTITY_SET_NAME =
14 | *     IV_SOURCE_NAME =
15 | *     IT_FILTER_SELECT_OPTIONS =
16 | *     IS_PAGING =
17 | *     IT_KEY_TAB =
18 | *     IT_NAVIGATION_PATH =
19 | *     IT_ORDER =
20 | *     IV_FILTER_STRING =
21 | *     IV_SEARCH_STRING =
22 | *     io_tech_request_context =
23 | ** IMPORTING
24 | *     et_entityset =
25 | *     es_response_context =
26 | *
27 | ** CATCH /iwbep/cx_mgw_busi_exception.
28 | ** CATCH /iwbep/cx_mgw_tech_exception.
29 | **ENDTRY.
30 | ENDMETHOD.
```

Now Go to Tcode : /N/IWFND/MAINT_SERVICE

Click Add Service

Activate and Maintain Services

Service Catalog

Type	Technical Service Name	Ver...	Service Description	External Service Name	Namespace	OAuth	Soft State	Status	Processing Mode
BEP	ZLAUNCHPADABAP_REPOSITORY_SRV	1	SAPUI5 ABAP Repository services via oData	ABAP_REPOSITORY_SRV	/UI5/	☐	Inactive		Routing-based
BEP	ZACCDT	1		ACCDT		☐	Not Supported		Co-deployed only
BEP	ADT	1	Gateway service for ADT	ADT_SRV		☒	Not Supported		Routing-based
BEP	ZANA_PAI_PS_SRV	1	Predictive Analytics Integrator (PAI) - PS Review	ANA_PAI_PS_SRV		☐			Co-deployed only
BEP	ZANA_PAI_REPOSITORY_SRV	1	Predictive Analytics Integrator (PAI) Repository	ANA_PAI_REPOSITORY_SRV		☐			Co-deployed only
BEP	ZANA_QUERY_DESIGNER	1	Analytical Query Designer	ANA_QUERY_DESIGNER		☐			Co-deployed only
BEP	ZAPI_BILLING_DOCUMENT_REQUEST_SRV	1	Remote API for SD Billing Document Request	API_BILLING_DOCUMENT_REQUEST_SRV		☒	Not Supported		Co-deployed only
BEP	ZAPI_BILLING_DOCUMENT_SRV	1	Remote API for SD Billing Document	API_BILLING_DOCUMENT_SRV		☐	Not Supported		Co-deployed only
BEP	/APPZEN/API_BUSINESS_PARTNER	1	Remote API for Business Partner	API_BUSINESS_PARTNER		☐	Not Supported		Co-deployed only
BEP	ZAPI_CHARTOFACCOUNTS_SRV	1	API for Chart of accounts	API_CHARTOFACCOUNTS_SRV		☐	Not Supported		Co-deployed only
BEP	/APPZEN/API_COMPANYCODE_SRV	1	API for CompanyCode	API_COMPANYCODE_SRV		☐	Not Supported		Co-deployed only
BEP	/APPZEN/API_COSTCENTER_SRV	1	API for Cost Center	API_COSTCENTER_SRV		☐	Not Supported		Co-deployed only
BEP	/APPZEN/API_CV_ATTACHMENT_SRV	1	Gateway: API for Attachment Service	API_CV_ATTACHMENT_SRV		☐	Not Supported		Co-deployed only
BEP	ZAPI_DISPUTECASEASCOORDTR	1	Dispute Case As Coordinator	API_DISPUTECASEASCOORDTR		☐			Co-deployed only
BEP	ZAPI_DISPUTECASEASPERSONRESP	1	Dispute Case As Person Responsible	API_DISPUTECASEASPERSONRESP		☐			Co-deployed only
BEP	ZAPI_DISPUTECASEASPROCESSOR	1	Dispute Case As Processor	API_DISPUTECASEASPROCESSOR		☐			Co-deployed only
BEP	/APPZEN/API_GLACCOUNTTOFACC_SRV	1	API for G/L Account	API_GLACCOUNTTOFACC_SRV		☐	Not Supported		Co-deployed only
BEP	/APPZEN/API_GLACCOUNTLINEITEM	1	Odata Service for A_GLAccountLineItem	API_GLACCOUNTLINEITEM		☐	Not Supported		Co-deployed only
BEP	/APPZEN/API_INTERNALORDER_SRV	1	API for Internal Order	API_INTERNALORDER_SRV		☐	Not Supported		Co-deployed only
BEP	ZAPI_JOURNALENTYITEMBASIC_SRV	1	Journal Entry Item - Read (A2X)	API_JOURNALENTYITEMBASIC_SRV		☐	Not Supported		Co-deployed only
BEP	ZAPI_MAINTNOTIFICATION	1		API_MAINTNOTIFICATION		☐			Co-deployed only

ICF Nodes

Status	ICF Node	Session Time-out	Soft State	Description
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System Aliases

SAP System Alias	Description	Default System	Metadata	User Role
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- 1- In System Alias: LOCAL
- 2- then Select your Odata Service Name
- 3- Click Add Selected Services

Add Selected Services

Get Services

Filter

System Alias: 

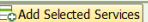
Technical Service Name:

External Service Name:

Version:

External Mapping ID:

☐ Co-Deployed

 Add Selected Services 

Select Backend Services

Type	Technical Service Name	Version	Service Description	External Service Name	Namespace
BEP	ZMASTERDATA_SRV	1	for excellon demo	ZMASTERDATA_SRV	
BEP	ZMATERIAL_DISPLAY_SRV	1	ZMATERIAL_DISPLAY	ZMATERIAL_DISPLAY_SRV	
BEP	ZMATERIAL_LIST_SRV	1	TEST	ZMATERIAL_LIST_SRV	
BEP	ZMATERIAL_LIST_SRV_01	1	TEST	ZMATERIAL_LIST_SRV_01	
BEP	ZMK_PRODUCT_SRV	1	product	ZMK_PRODUCT_SRV	
BEP	ZOD_08_SRV	1	ODATA SERVICE	ZOD_08_SRV	
BEP	ZOD_GW_68_SRV	1	Odata service tom fetch material info	ZOD_GW_68_SRV	
BEP	ZOD_GW08_SRV	1	ODATA SERVICE TO FETCH INFO	ZOD_GW08_SRV	
BEP	ZOD_GW121_SRV	1	THAT CRAETE THE	ZOD_GW121_SRV	
BEP	ZODATA_01_SRV	1	odata services	ZODATA_01_SRV	
BEP	ZODATA_02_SRV	1	odata service	ZODATA_02_SRV	
BEP	ZODATA_09_SRV	1	odata	ZODATA_09_SRV	
BEP	ZODATA_11_SRV	1	odata	ZODATA_11_SRV	
BEP	ZODATA_72_SRV	1	odata service	ZODATA_72_SRV	
BEP	ZODATA_AT_1_SRV	1	api	ZODATA_AT_1_SRV	
BEP	ZODATA_MAKTX_SRV	1	material description	ZODATA_MAKTX_SRV	
BEP	ZODATA_SERVICE_1_SRV	1	odata service	ZODATA_SERVICE_1_SRV	
BEP	ZODATA_SERVICE_11_SRV	1	ODATA	ZODATA_SERVICE_11_SRV	
BEP	ZODATA_SERVICE_12_SRV	1	ODATA SERVICE	ZODATA_SERVICE_12_SRV	
BEP	ZPACK_INITIATE_SRV	1	PACK INITIATE	ZPACK_INITIATE_SRV	
BEP	ZPO_102594_SRV	1	PURCHASE ORDER ODATA	ZPO_102594_SRV	
BEP	ZPRACTICF_ODATA_SRV	1	test	ZPRACTICF_ODATA_SRV	

Continue the process

Add Service

Service

Technical Service Name:

Service Version:

Description:

External Service Name:

Namespace:

External Mapping ID:

External Data Source Type:

Model

Technical Model Name:

Model Version:

Creation Information

Package Assignment:

ICF Node

☒ Standard Mode ☐ None

☒ Set Current Client as Default Client in ICF Node

OAuth enablement

☐ Enable OAuth for Service

Go back to Tcode /N/IWFND/MAINT_SERVICE

1. find your Odata Service and select it
2. Click "SAP Gateway Client "

Activate and Maintain Services

Service Catalog

Type	Technical Service Name	Ver...	Service Description	External Service Name	Namespace	OAuth ...	Soft Stat
BEP	ZMK_SALES_SRV	1	sales order data	ZMK_SALES_SRV		☐	Not Supp
BEP	ZMK_SALESDATA_SRV	1	sales data	ZMK_SALESDATA_SRV		☐	Not Supp
BEP	ZMK_SALESNEW_SRV	1	sales	ZMK_SALESNEW_SRV		☐	Not Supp
BEP	ZMKHEADERANDITEM_SRV	1	HEADERANDITEM	ZMKHEADERANDITEM_SRV		☐	Not Supp
BEP	ZMOHIT_PHONE_SRV	1	About my Phone	ZMOHIT_PHONE_SRV		☐	Not Supp
BEP	ZMOLI_ODATA1_SRV	1	odata srv1	ZMOLI_ODATA1_SRV		☒	Not Supp
BEP	ZMY_ODATA_PROJECT_SRV	1	MY_ODATA_PROJECT	ZMY_ODATA_PROJECT_SRV		☐	Not Supp
BEP	ZOCT_ODATA_SRV	1	October OData	ZOCT_ODATA_SRV		☐	Not Supp
BEP	ZOD_GW_68_SRV	1	Odata service tom fetch material info	ZOD_GW_68_SRV		☐	Not Supp
BEP	ZOD_GW_OM_SRV	1	ODATA SERVICE TO MATERIAL INFO	ZOD_GW_OM_SRV		☐	Not Supp
BEP	ZOD_GW1_SRV	1	Odata service to fetch material info	ZOD_GW1_SRV		☐	Not Supp
BEP	ZOD_GW11_SRV	1	OData Service to fetch Material Info	ZOD_GW11_SRV		☐	Not Supp
BEP	ZODATA_07_SRV	1	Odata Services	ZODATA_07_SRV		☐	Not Supp
BEP	ZODATA_22_SRV	1	DEMO	ZODATA_22_SRV		☐	Not Supp
BEP	ZODATA_23_SRV	1	ODATA SERVICE	ZODATA_23_SRV		☒	Not Supp
BEP	ZODATA_24_SRV	1	ODATA SERVICE	ZODATA_24_SRV		☒	Not Supp
BEP	ZODATA_AIRLINE_PRO_SRV	1	for o data practise	ZODATA_AIRLINE_PRO_SRV		☐	Not Supp
BEP	ZODATA_AIRLINE_PROJECT_SRV	1	for airline o data practise	ZODATA_AIRLINE_PROJECT_SRV		☐	Not Supp
BEP	ZODATA_MAKTX_PJ_SRV	1	Material Description	ZODATA_MAKTX_PJ_SRV		☐	Not Supp
BEP	ZODATA_PURCHASE_DETAILS_SRV	1	Project of OData Purchase details	ZODATA_PURCHASE_DETAILS_SRV		☐	Not Supp
BEP	ZODATA_RAP_SRV	1	PRACTICE	ZODATA_RAP_SRV		☒	Not Supp

ICF Nodes

Status	ICF Node	Session Time-out	Soft State	Description
	ODATA	00:00:00		Standard Mode

System Aliases

SAP System Alias	Description
LOCAL	Local System Alias

ICF Node: **SAP Gateway Client**

Open Entity Sets and choose yours

Note (you can have more than entity Set in your project, each one have different structure)

SAP Gateway Client

Execute Select Service Administration Service Implementation Switch User Entity Set Add URI Option

HTTP Method: ☒ GET ☐ POST ☐ PUT ☐ PATCH ☐ MERGE ☐ DELETE ☐ HEAD

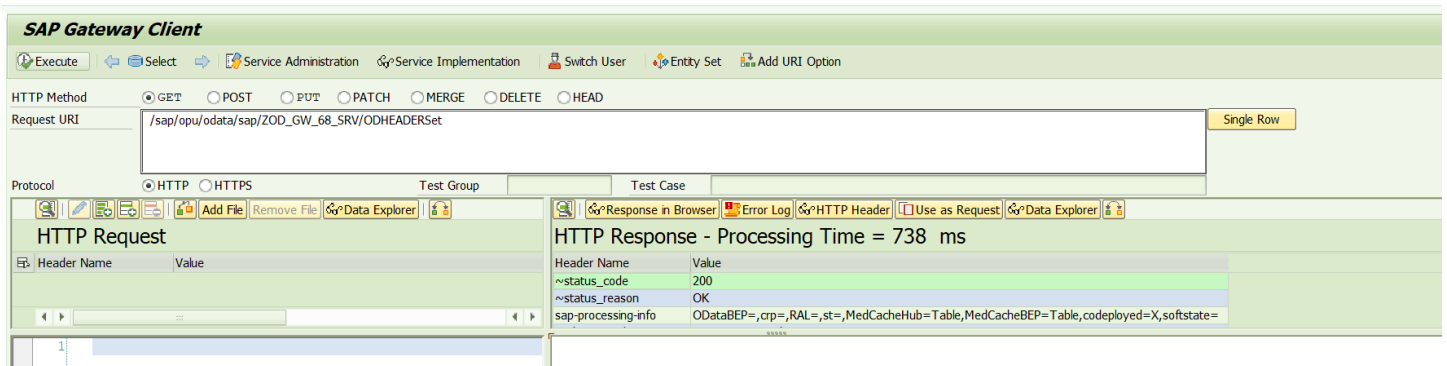
Request URI: Single Row

Protocol: ☒ HTTP ☐ HTTPS

Test Group: Test Case:

After select your EntitySet.

Click Execute to test it.



Once the service is running , you can test it in a browser or use tools like Postman.

The OData service can be consumed by external applications and SAP Fiori applications using standard OData protocols.